

SaaS Growth Economics

Assessing ARR growth, retention, and customer acquisition economics

Self-directed portfolio project using synthetic data

BUSINESS QUESTION
Is ARR growth sustainable given current retention, customer mix, and acquisition economics?

ANALYSIS WINDOW Apr 2024 – Sep 2025	AS-OF DATE 2025-09-30	CUSTOMERS 100 synthetic logos	ARR BASIS Run-rate (MRR × 12)
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KEY FINDING
ARR peaked at \$73,560 in Jul 2025 and declined approximately 19% to \$59,400 by Sep 2025. Growth depended on new customers and reactivations; the data includes no upgrade-driven expansion revenue.

\$73,560
peak run-rate ARR in Jul 2025

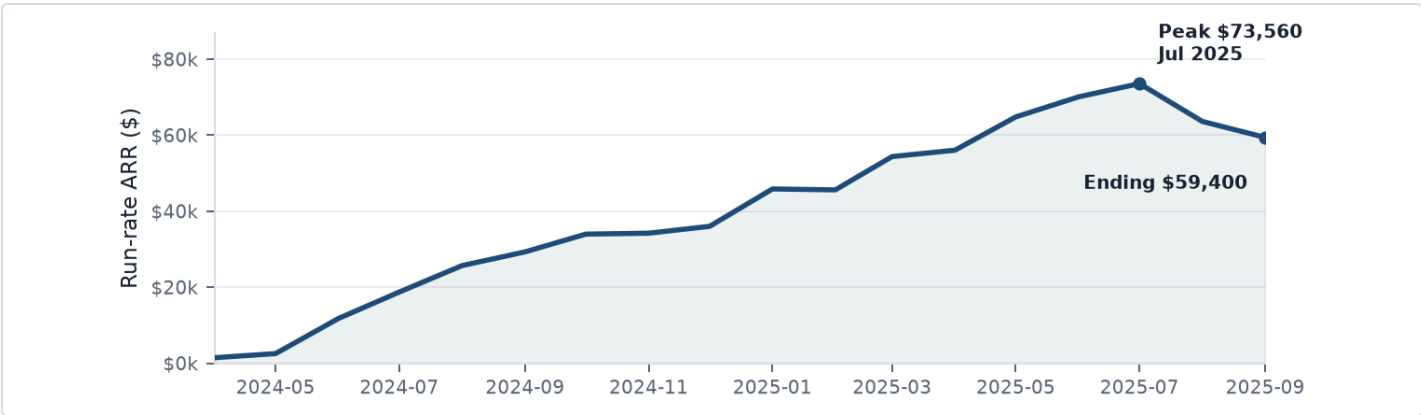


Figure 1. Run-rate ARR. Peak \$73,560 in Jul 2025; ending \$59,400 (\$4,950 MRR).



Figure 2. ARR bridge, April–September 2025. Expansion and contraction are zero because the data contains no upgrades or downgrades. Reactivation is presented separately from expansion.

FP&A assessment: Growth relied primarily on new-customer acquisition. From April through September 2025, \$32.9k of new ARR was offset by \$32.9k of churn ARR; \$5.0k of reactivation produced the net increase. Without stronger retention or expansion revenue, maintaining ARR will require continued customer acquisition.

Retention does not support growth without continued acquisition

Performance of the beginning recurring-revenue base

NRR and GRR are identical because the data contains no upgrades or downgrades; churn is the only movement within the beginning recurring-revenue base. New customers and reactivations are excluded from both measures.

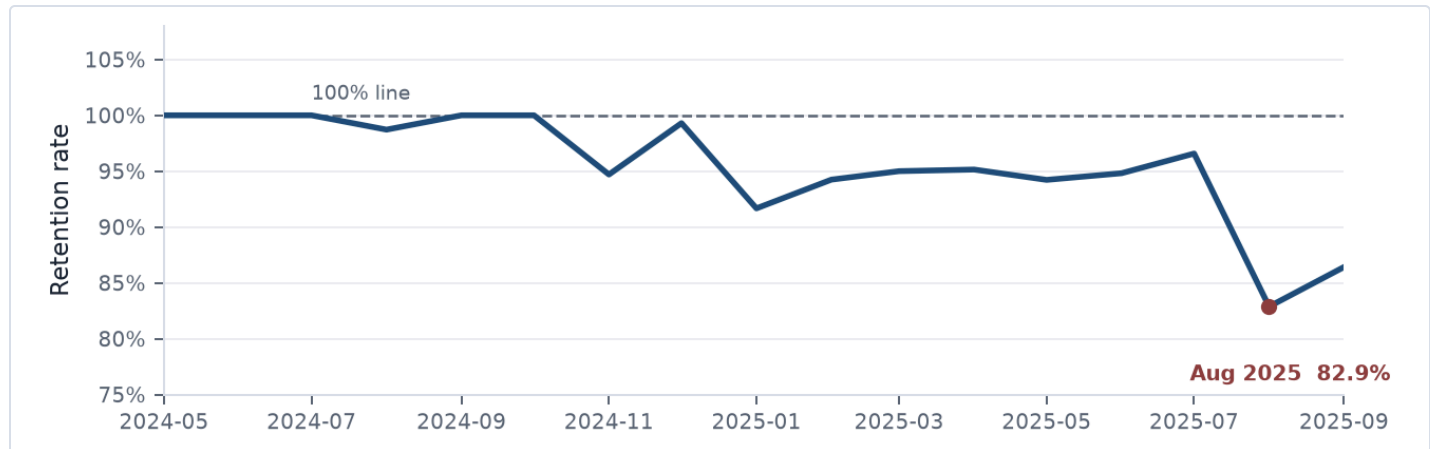


Figure 3. Monthly NRR / GRR. Mean 95.5%. August 2025: 82.9%. Months with NRR above 100%: 0.

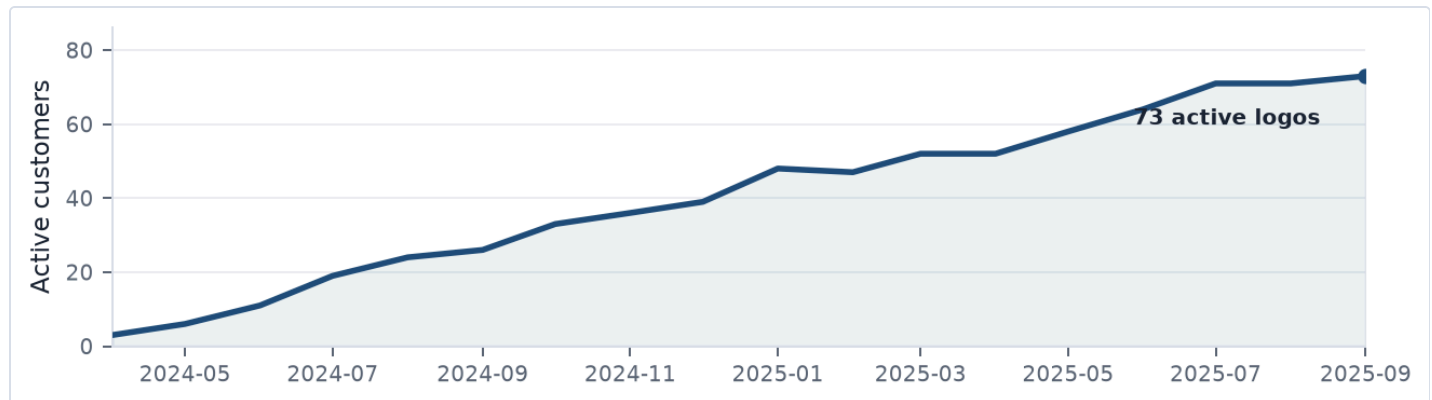


Figure 4. Active customers rose to 73 by Sep 2025 even as ARR fell from the July peak. Logo growth did not prevent a revenue decline.

Risks

- Mean NRR of 95.5% indicates that the beginning recurring-revenue base contracts without new customers or reactivation.
- August NRR of 82.9% shows how quickly elevated churn can reduce recurring revenue.
- A growth plan that depends on acquisition must continually replace churned ARR, increasing the importance of acquisition efficiency.

Opportunities

- Developing an expansion and upsell motion could improve NRR if product adoption and customer demand support it.
- Segment churn by channel, customer segment, cohort, and tenure before increasing acquisition spending.
- Reactivation provides some recovery, but it does not replace stronger retention or expansion.

Method note: NRR and GRR measure performance of the beginning recurring-revenue base and exclude new customers and reactivations. Signup-cohort dollar retention is not shown because later subscription starts and reactivations can push it above 100%, making it unsuitable as an NRR measure.

Acquisition efficiency requires tighter investment discipline

Estimated CAC payback = assigned CAC ÷ (ARPU × 80% gross margin)

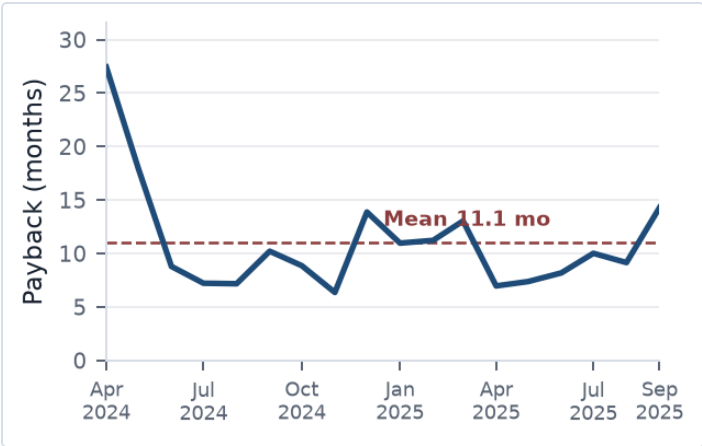


Figure 5. Estimated CAC payback. Mean 11.1 months. CAC is based on assigned customer acquisition cost, not GL marketing spend.

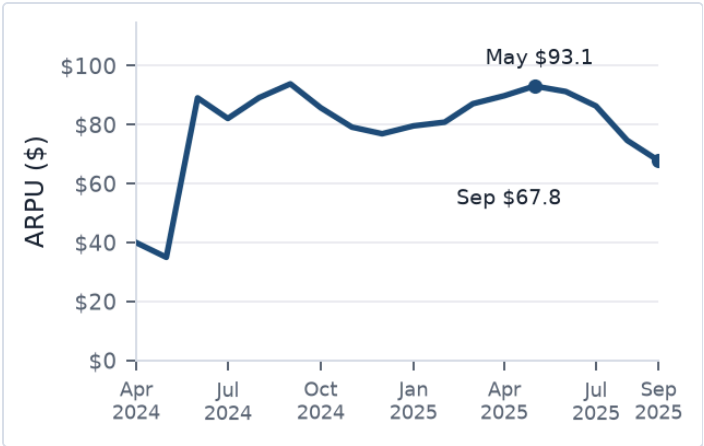


Figure 6. Blended ARPU fell from \$93.1 in May 2025 to \$67.8 in Sep 2025, reducing the contribution available to recover CAC.

Channel snapshot

CHANNEL	NEW LOGOS	AVG ASSIGNED CAC	SEP ARPU	EST. PAYBACK
Paid	39	\$931	\$61	19.2 mo
Organic	31	\$213	\$94	2.8 mo
Partner	20	\$732	\$58	15.9 mo
Outbound	10	\$560	\$37	18.8 mo

Estimated payback = average assigned CAC ÷ (September channel ARPU × 80% gross margin). Use this measure for relative channel comparison, not as a vintage payback curve. Partner and Outbound have small samples.

Recommended actions

1. Separate the forecast into beginning-base retention, new customers, reactivation, and expansion rather than applying a single growth rate.
2. Analyze churn by customer segment, channel, cohort, and tenure before increasing acquisition spending.
3. Add an explicit expansion assumption to the forecast instead of relying on new customers to replace churned ARR.
4. Update CAC payback guardrails as ARPU, gross margin, and retention change; do not rely on a static portfolio average.

Management implications

- ARR growth through July masked the absence of upgrade-driven expansion revenue.
- Declining ARPU increases CAC payback even when acquisition cost is unchanged.
- Paid acquisition is the largest source of new customers and has the longest estimated payback in the current channel mix.

Limitations

- Synthetic data; results do not represent a real company.
- Analysis is frozen through September 30, 2025.
- CAC uses assigned customer acquisition cost rather than GL marketing-spend allocation.
- The data contains no upgrade or downgrade activity.
- No causal inference or production deployment.